

Quiz 3. Let h be an unkeyed compression function that takes a 256-bit chaining value and a 512-bit message block and returns a 256-bit chaining value. Let H be the hash function built from h via the Merkle-Damgård transform with strengthening.

HMAC is a keyed hash function built from H . For a key $K \in \{0, 1\}^{256}$, define

$$K_i = (K \| 0^{256}) \oplus \textit{ipad} , \quad K_o = (K \| 0^{256}) \oplus \textit{opad} ,$$

where $\textit{ipad}$ and $\textit{opad}$ are fixed 512-bit constants, so K_i and K_o are each exactly one 512-bit block. Then

$$\text{HMAC}_K(M) = H(K_o \| H(K_i \| M)) .$$

Question. How many compression-function calls are made in total to compute $\text{HMAC}_K(M)$ for a message $M \in \{0, 1\}^{1024}$? Show your work.